Q1: Where is DDR register located (for each port)?

The DDR registers (DDRA, DDRB, DDRC, DDRD) are located in the **I/O Register Space of the Data Memory** in the Atmega32

Port A Data Direction Register – DDRA

Bit	7	6	5	4	3	2	1	0	
	DDA7	DDA6	DDA5	DDA4	DDA3	DDA2	DDA1	DDA0	DDRA
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Port C Data Direction Register – DDRC

Bit	7	6	5	4	3	2	1	0	
	DDC7	DDC6	DDC5	DDC4	DDC3	DDC2	DDC1	DDC0	DDRC
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Port D Data Direction Register – DDRD

Bit	7	6	5	4	3	2	1	0	
	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Port B Data Direction Register – DDRB

Bit	7	6	5	4	3	2	1	0	
	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	DDRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Q2: SRAM Vs Flash?

- SRAM: Part of the **Data Memory**
- Stores **variables during program execution**
- **Volatile** → data is lost when power is off

SRAM Data Memory

Figure 9 shows how the ATmega32 SRAM Memory is organized.

The lower 2144 Data Memory locations address the Register File, the I/O Memory, and the internal data SRAM. The first 96 locations address the Register File and I/O Memory, and the next 2048 locations address the internal data SRAM.

The five different addressing modes for the data memory cover: Direct, Indirect with Displacement, Indirect, Indirect with Pre-decrement, and Indirect with Post-increment. In the Register File, registers R26 to R31 feature the indirect Addressing Pointer Registers.

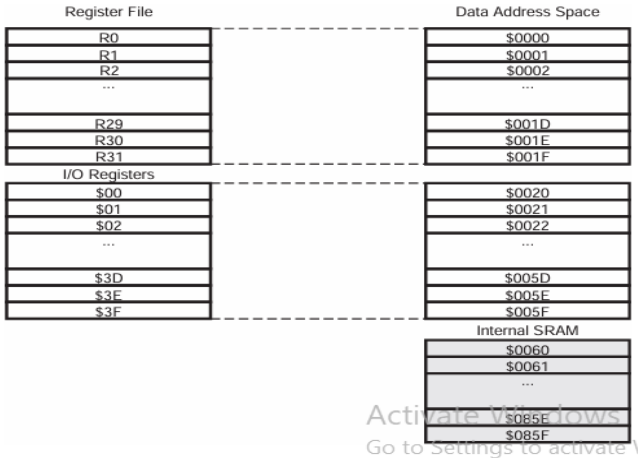
The direct addressing reaches the entire data space.

The Indirect with Displacement mode reaches 63 address locations from the base address given by the Y- or Z-register.

When using register indirect addressing modes with automatic pre-decrement and post-increment, the address registers X, Y, and Z are decremented or incremented.

The 32 general purpose working registers, 64 I/O Registers, and the 2048 bytes of internal data SRAM in the ATmega32 are all accessible through all these addressing modes. The Register File is described in "General Purpose Register File" on page 11.

Figure 9. Data Memory Map



Activate Windows
Go to Settings to activate Windows.

- FLASH: Stores the **Program Code**
- **Non-volatile** → data remains after power is off
- Larger than SRAM in the microcontroller

In-System Reprogrammable Flash Program Memory

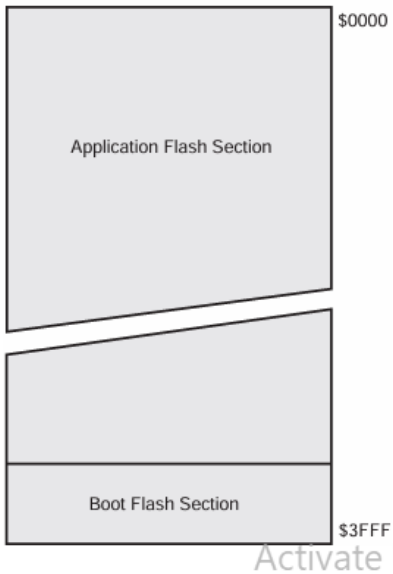
The ATmega32 contains 32K bytes On-chip In-System Reprogrammable Flash memory for program storage. Since all AVR instructions are 16 or 32 bits wide, the Flash is organized as 16K x 16. For software security, the Flash Program memory space is divided into two sections, Boot Program section and Application Program section.

The Flash memory has an endurance of at least 10,000 write/erase cycles. The ATmega32 Program Counter (PC) is 14 bits wide, thus addressing the 16K program memory locations. The operation of Boot Program section and associated Boot Lock bits for software protection are described in detail in "Boot Loader Support – Read-While-Write Self-Programming" on page 244. "Memory Programming" on page 256 contains a detailed description on Flash Programming in SPI, JTAG, or Parallell Programming mode.

Constant tables can be allocated within the entire program memory address space (see the LPM – Load Program Memory Instruction Description).

Timing diagrams for instruction fetch and execution are presented in "Instruction Execution Timing" on page 13.

Figure 8. Program Memory Map



Activate Windows